assign points with the same x coordinate of our pivot to the left or right partition, but we'll have to do so for all of them, no matter how far they are from 0 on the y axis.

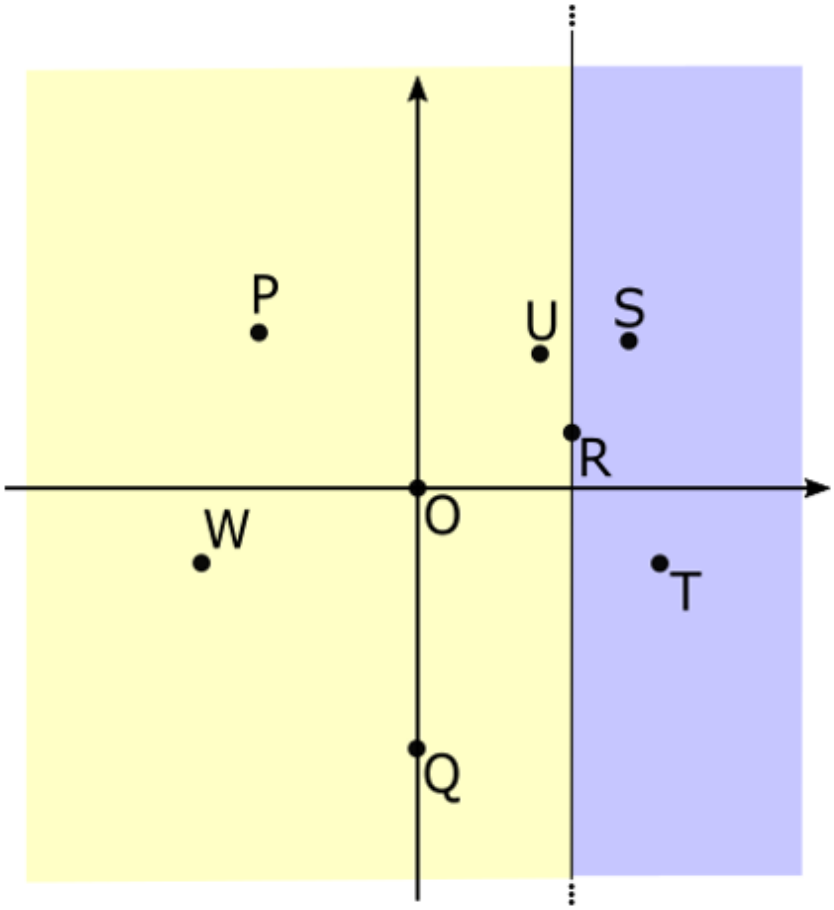


Figure 8.9 Another example of partitioning $\mathbb{R}^2$ space using a line parallel to y axis and passing for a given point (R, in this case).

Both examples show issues that derive from the same mistake: we are ignoring the points' y coordinates altogether. Using only a fraction of the available information can't be ideal: whenever we give up some info about a dataset, we